OPERATION

2.4L

The Camshaft Position Sensor (CMP) sensor contains a hall effect device referred to as a sync signal generator. A rotating target wheel (tonewheel) for the CMP is located behind the exhaust valve-camshaft drive gear (Fig. 7). The target wheel is equipped with a cutout (notch) around 180 degrees of the wheel. The CMP detects this cutout every 180 degrees of camshaft gear rotation. Its signal is used in conjunction with the Crankshaft Position Sensor (CKP) to differentiate between fuel injection and spark events. It is also used to synchronize the fuel injectors with their respective cylinders.

When the leading edge of the target wheel cutout enters the tip of the CMP, the interruption of magnetic field causes the voltage to switch high, resulting in a sync signal of approximately 5 volts.

When the trailing edge of the target wheel cutout leaves the tip of the CMP, the change of the magnetic field causes the sync signal voltage to switch low to 0 volts.

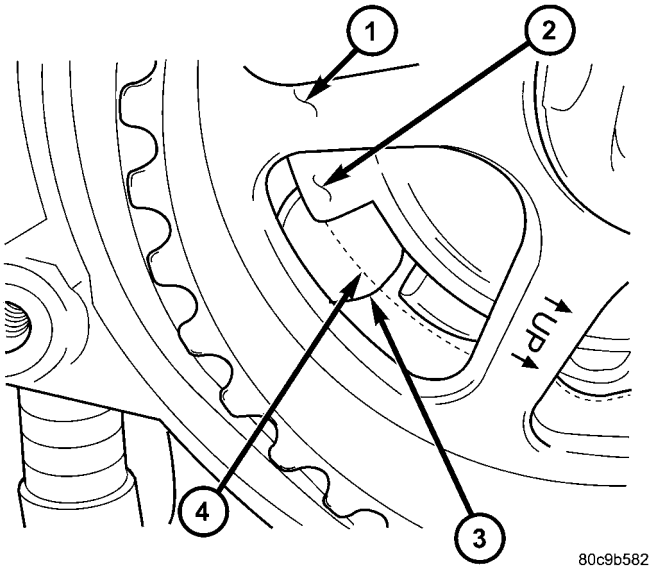

Fig. 7 CMP FACE AT TARGET WHEEL-2.4L

1 - CAMSHAFT DRIVE GEAR
2 - TARGETWHEEL (TONEWHEEL)
3 - FACE OF CMP SENSOR
4 - CUTOUT (NOTCH)

4.0L

The CMP sensor (1) (Fig. 8) uses a rotating hall effect device called a target wheel (3) which is attached to the oil pump drive shaft. The target wheel contains sets of machined notches (2).

When the leading edge of a target wheel notch passes by the tip of the CMP sensor, the interruption of magnetic field causes the voltage to switch high resulting in a sync signal of approximately 5 volts.

When the trailing edge of the target wheel notch passes by the tip of the CMP sensor, the change of the magnetic field causes the sync signal voltage to switch low to 0 volts.

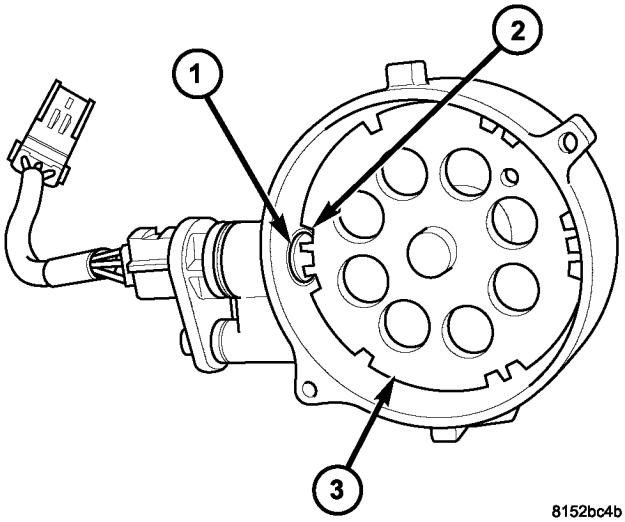

Fig. 8 4.0L CMP OPERATION

1 - CMP SENSOR
2 - TARGET WHEEL NOTCHES
3 - TARGET WHEEL

REMOVAL

2.4L

The Camshaft Position Sensor (CMP) on the 2.4L 4–cylinder engine is bolted to the right-front side of the cylinder head (Fig. 9). Sensor position (depth) is adjustable.

(1) Disconnect electrical connector at CMP sensor.
(2) Remove 2 sensor mounting bolts.
(3) Remove sensor from cylinder head by sliding towards rear of engine.